Reading the Reading Room

- Pneumonia classification from pediatric chest X-rays
- Final milestones: anatomy-guided inputs and matched classifier comparisons
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From reproduction to a controlled experiment

- M1: literature review and dataset selection
- M2: reproduced five-CNN baseline and audited predictions
- M3: predicted gains from lung cropping and a pretrained transformer
- M4–M5: implementation, measured comparisons and failure analysis

Dataset and evaluation

- Normal versus pneumonia on the Kermany pediatric benchmark
- Historical baseline: 5,232 training records and 624 test records
- New split: 4,164 training, 1,042 validation and 624 test records
- The benchmark test was inspected previously. It is not an external blind holdout.

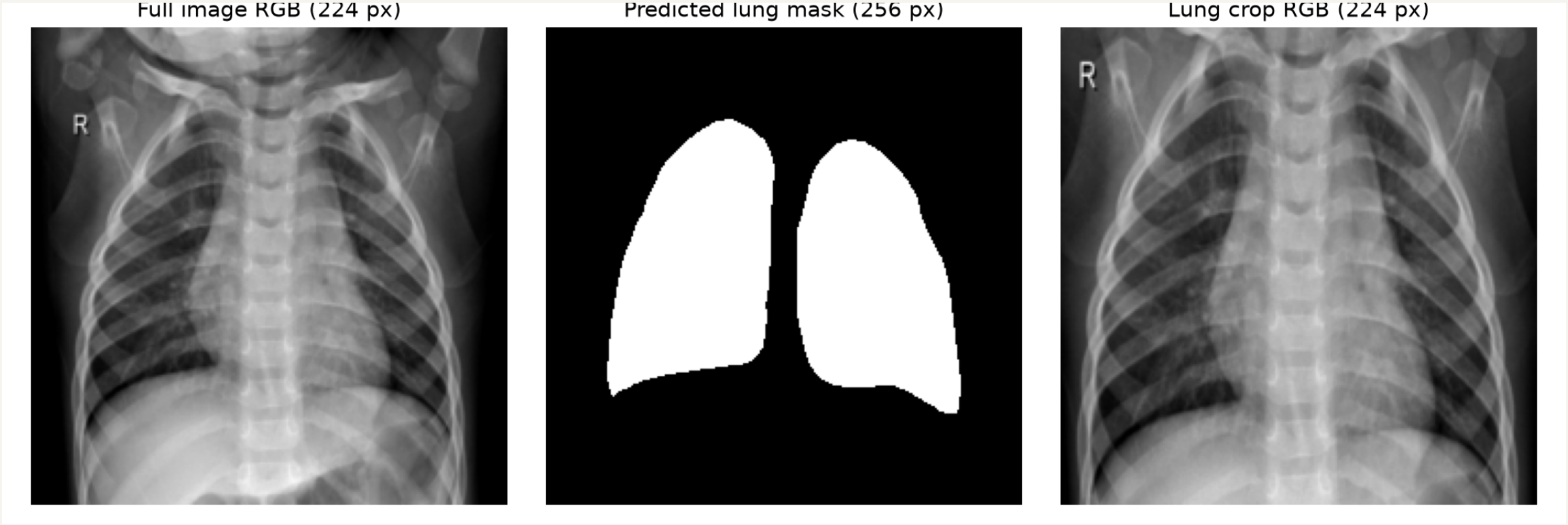
The verified M2 baseline

- Five pretrained CNNs, five epochs, majority voting. Paper target: 96.4% accuracy.
- Accuracy 92.15% · pneumonia recall 94.62% · specificity 88.03%
- Macro F1 0.916 · ROC AUC 0.962
- 28 false positives, 21 false negatives. ResNet18 alone: 93.11% accuracy.

Why focus on the lungs?

- The full image includes peripheral anatomy and acquisition markings
- A fixed pretrained anatomical model predicts both lungs at 256 pixels
- We crop their union with a 10% margin, then resize for classification
- 256 replaces the standard 512 input for CPU runtime. Failed masks fall back.
- Median retained crop area: 71.1% of original image area

The crop pipeline on a training image



The implementation and its limits

- Frozen pretrained image encoders with five-epoch trained classifier heads
- ResNet18 supplies CNN features. DeiT-Tiny supplies transformer features.
- The pilot isolates image region and classifier combination under one split
- This reduced experiment differs from the original five-CNN late-block proposal.
- Uploaded 224 px fixture: 4.58 s including segmentation; cached cases: 1.24–1.49 s.

Four matched conditions

- A: full image with ResNet18
- B: lung-focused crop with ResNet18
- C: full image with ResNet18 and DeiT-Tiny
- D: lung-focused crop with ResNet18 and DeiT-Tiny
- Checkpoints and combination weights use validation data only.

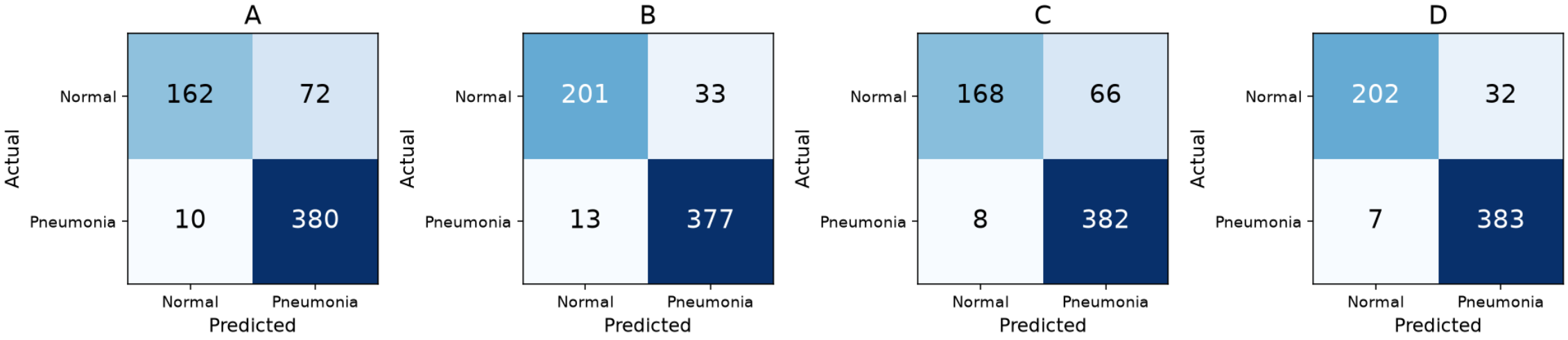
Measured results

Run	Accuracy	Recall	Specificity	Macro F1	AUC
A	86.86%	97.44%	69.23%	0.850	0.931
B	92.63%	96.67%	85.90%	0.920	0.970
C	88.14%	97.95%	71.79%	0.866	0.947
D	**93.75%**	**98.21%**	**86.32%**	**0.932**	**0.976**

A: full CNN. B: crop CNN. C: full CNN + ViT. D: crop CNN + ViT.

624 test records. Validation-selected ViT weights: C 0.25; D 0.50.

Confusion matrices by condition



What happened to the M3 prediction?

- Crop effect B minus A: macro F1 +0.0696, specificity +16.67 percentage points
- Paired 95% interval for macro-F1 difference: [0.0404, 0.1016]
- The measured macro-F1 gain exceeds the predicted +0.03 upper bound.
- Validation-selected ViT weights: C=0.25, D=0.5

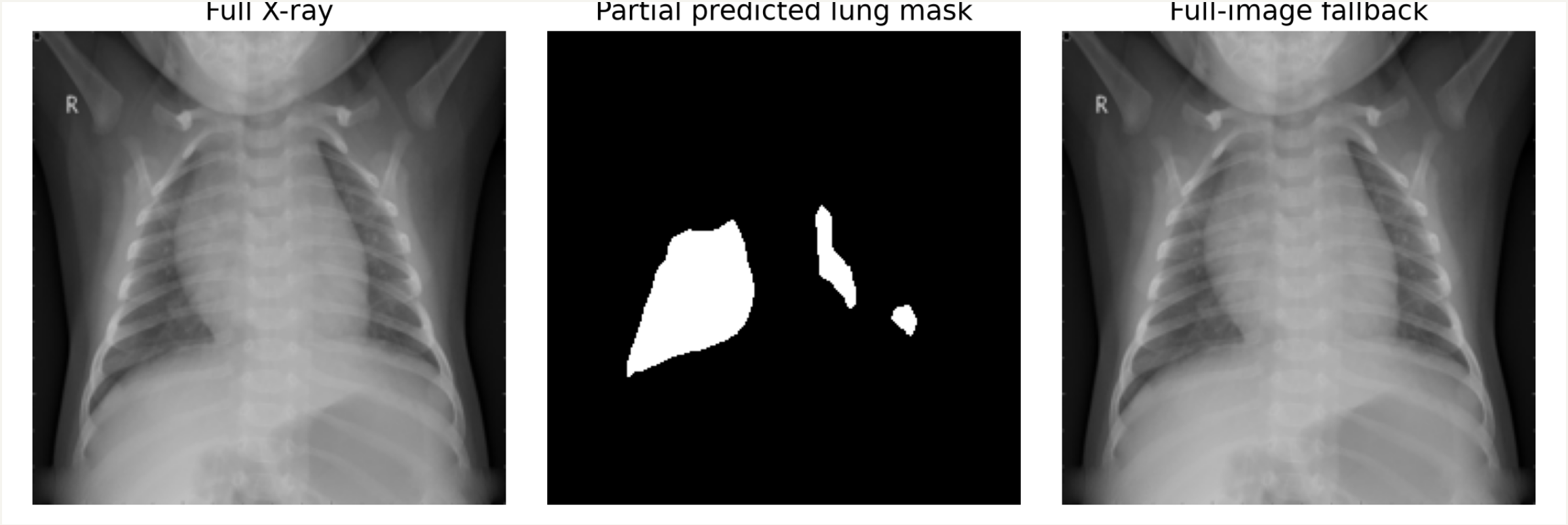
Validation and test disagree on cropping

- A: validation macro F1 0.955; test 0.850
- B: validation macro F1 0.907; test 0.920
- C: validation macro F1 0.957; test 0.866
- D: validation macro F1 0.924; test 0.932
- This single split supports a benchmark result, not universal crop superiority.

Errors and uncertainty

- Automatic full-image fallbacks: 82 / 5824 (1.41%)
- Inspect false positives, false negatives and segmentation failures
- Report sensitivity alongside specificity and macro F1
- Paired uncertainty estimates describe this benchmark, not other hospitals
- Grad-CAM highlights one CNN response. It does not establish lesion boundaries.

When the pipeline falls back to the full image



Live demonstration

- Select a benchmark example or upload a PNG/JPEG
- Inspect the model prediction, individual scores and Grad-CAM
- Inspect lung crop experiment metrics and recorded protocol
- The interface identifies the model being demonstrated and uses saved trained weights.

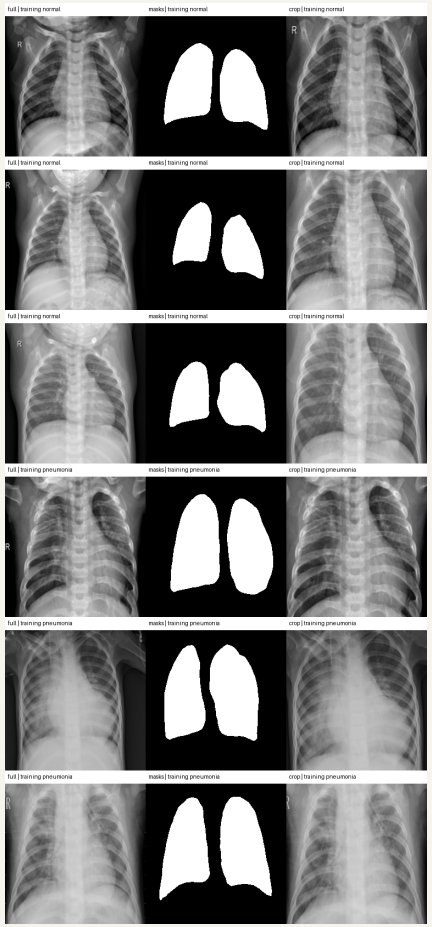
Conclusions and remaining questions

- Predeclared crop CNN–ViT: 93.75% accuracy and 98.21% pneumonia recall
- Small pediatric data and imperfect identifier groups limit generalization
- Adding ViT to the crop CNN improved accuracy by 1.12 percentage points
- The validation/test reversal requires repeated splits and external testing.

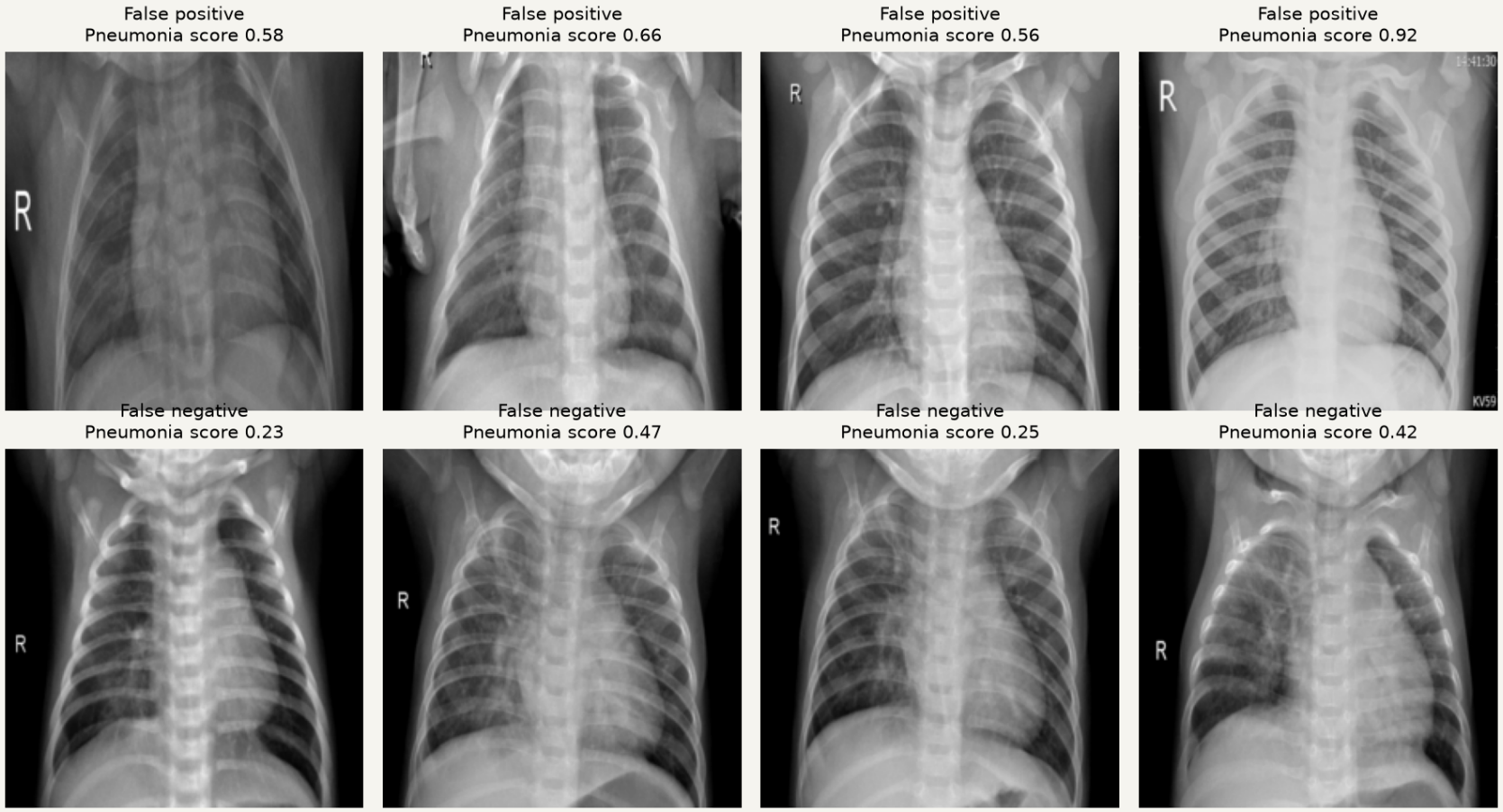
Appendix: Interpreting the comparison

- B versus A tests the crop effect for the CNN pilot
- C versus A tests the transformer addition on full images
- D versus B tests the transformer addition on crops
- The historical M2 score uses a different training protocol and is context only.

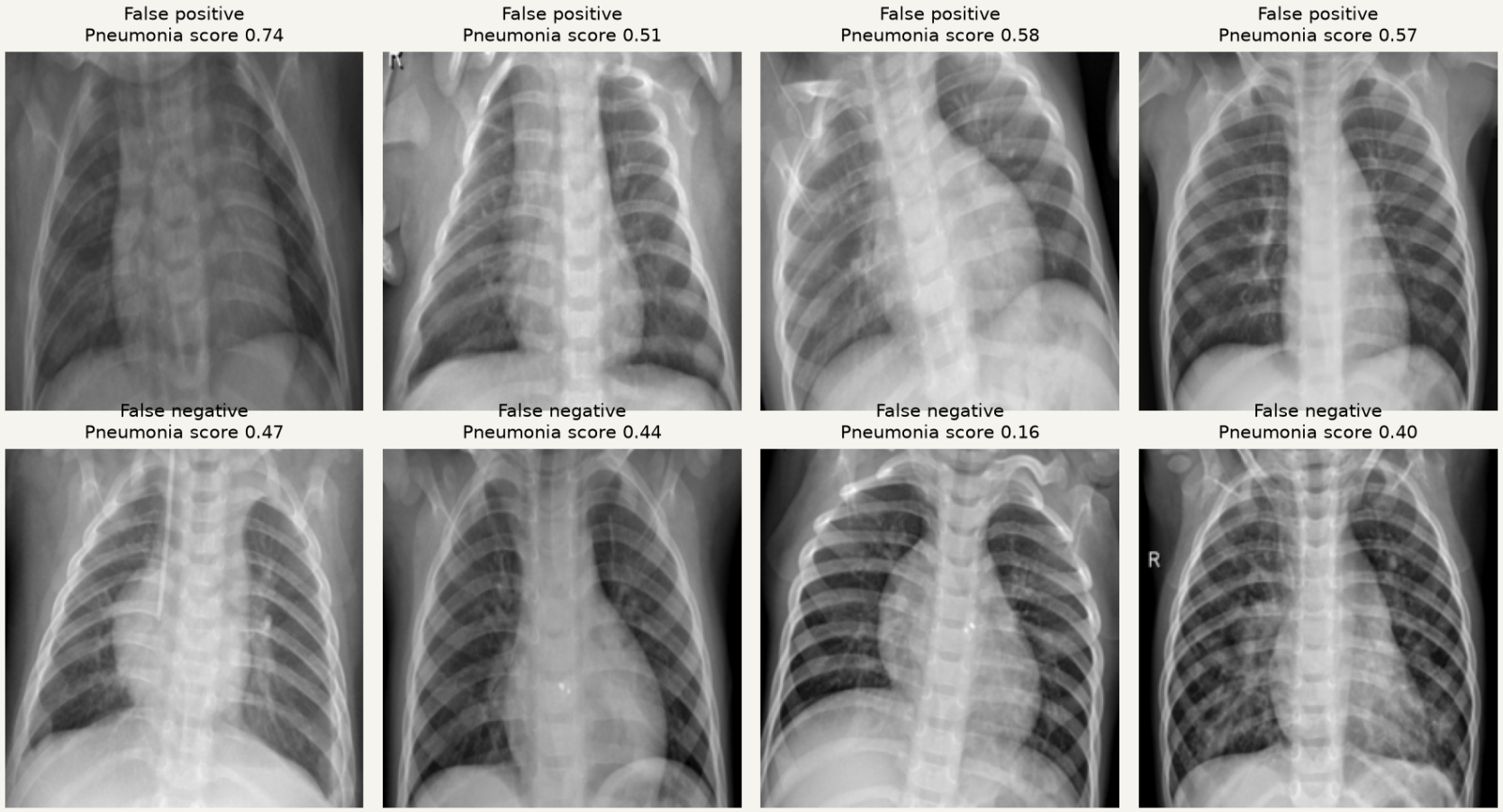
Appendix: Training examples from both classes



Appendix: Historical baseline error examples



Appendix: Where the crop hybrid makes errors



References

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